

Masters Comprehensive Examination

Applied (90 Minutes)

March 21, 2009

Answer **ONLY 3 QUESTIONS**. Open book and open notes.

1. Weights of pears are normally distributed. Dr. Y believes that the average weight of a pear grown in Hawaii should be exactly the same as the average weight of a pear grown in Michigan. He has recorded data from

$$n = 11 \text{ pears in Hawaii with } \bar{X}_H = 1030.5 \text{ gm } S_H^2 = 500.1$$

$$n = 11 \text{ pears in Michigan with } \bar{X}_M = 1029.3 \text{ gm } S_M^2 = 1000.1$$

A. Test $H_0: \sigma_H^2 = \sigma_M^2$ against $H_0: \sigma_H^2 \neq \sigma_M^2$ at $\alpha=0.10$. Show your work and give the result

B. Dr. X is certain that the null hypothesis $\mu_H = \mu_M$ is True. But he also believes that Dr. Y has falsified his data to make the results better than they should be by chance alone. He wants you to examine Dr. X's data for any evidence of this type of falsification.

- i) Formulate and state null and alternate hypotheses to test this. You can use words instead of symbols to do this if you wish, keep your answer reasonably short.
- ii) Test the null hypothesis at $\alpha=0.05$. For this question, now assume that σ^2 is known and equals 750 for both pears grown in Hawaii and in Michigan

2. The time it takes for an engine to fail is exponentially distributed. We have observed the following ordered failure times (in years) for 8 engines

1.2 2.3 4.3 5.4 7.6 8.4 10.2 14.5

A. Give the best estimate of the mean failure time and the standard deviation of this estimate and state all assumptions needed.

B. Give the best estimate of the timepoint X for which exactly 90% of the engines fail in X or less years. Give a 95% confidence interval for X.

3. Unborn babies of pregnant woman infected with HIV, are at risk at contracting HIV at birth. A doctor believes when there are twins, the one who is born first has a bigger chance to be HIV infected than does the one who is born second. He has data from 200 mothers infected with HIV that had twins; a) for 50 of these mothers both twins were born infected with HIV; b) for 20 mothers the first born twin was infected with HIV and the second born twin was not; c) for 10 the second born twin was infected with HIV and the first born was not and, d) for 120 mothers neither twin was infected.

A. Use a large sample approximation to derive the best test statistic for the null hypothesis that the first and second born twins are equally likely to be HIV infected and compare this to the rejection region at a two sided $\alpha = 0.05$.

B. Based on this data if a randomly chosen twin born to an HIV infected mother is infected with HIV, give a point estimate and 95% confidence interval for the probability that the sibling twin (the chosen twin's twin brother or sister) is also infected with HIV **Assume that the order of birth does not matter for this problem.**

4. Two methods (Method A and Method B) to produce vials of a vaccine are tested with the outcome being that the vial is either good or bad. Each method is allowed to produce new vials one at a time until the first bad vial from that method was observed, then production using that method is stopped. In other words,

- For Methods A & B each, if the last vial produced is good the method is allowed to produce another vial.
- But once a bad vial is produced, the Method is stopped and no new vials are produced with that method.

A. Assume that the probability for a vial produced by Method A to be bad is $p = 0.05$ and the probability for a vial produced by Method B to be bad is 0.10 and all vials are independent.

- i) What is the expected total number of good vials produced by both Methods A and B combined before both methods are stopped?
- ii) What is the variance of the total number of good vials produced by both Methods A and B combined before both methods are stopped?

B. Now assume that the probabilities for vials produced by each method to be bad are unknown and the following results are observed:

- Method A - On its first try, the vial was bad so production stopped
- Method B produced 15 good vials but on its 16th try, the vial was bad so production was stopped then

Derive the appropriate model that conditions on 15 bad vials being produced and give a P-value for observing a result this extreme if each method is equally likely to produce a bad vial.
